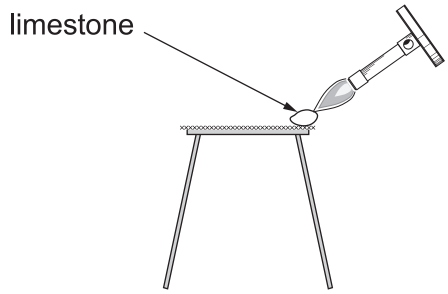
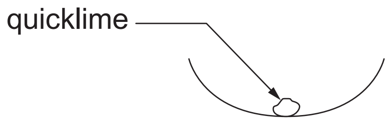
7. The flow chart shows the stages a student carried out to change calcium carbonate (limestone) into calcium hydroxide solution (limewater).

Stage 1



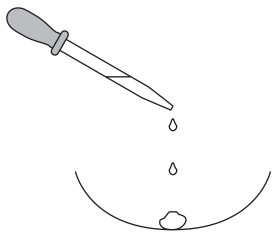
Limestone is roasted for 15 minutes to make quicklime

Stage 2



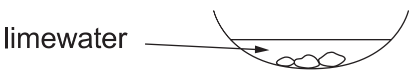
Quicklime is cooled for 20 minutes

Stage 3



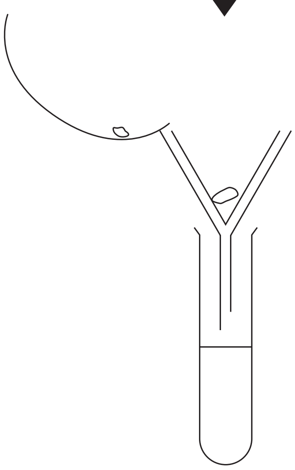
A few drops of water are added to the cold quicklime to make slaked lime

Stage 4



Excess water is added to the slaked lime to make limewater

Stage 5



Limewater is filtered

Examiner only

(a) Use the information in the flow chart to answer parts (i) to (iii).

(i) Name the type of reaction taking place in stage **1**. [1]

....

(ii) Give the **number** of the stage which shows an exothermic reaction. [1]

....

(iii) I. Describe what you would expect to **see** happen in stage **3**. [2]

....

....

....

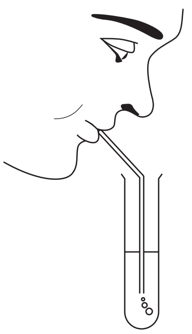
II. The reaction taking place in stage **3** is shown by the following word equation.

calcium oxide + water $\longrightarrow$ calcium hydroxide

Write a balanced **symbol** equation for this reaction. [2]

.... + ... $\longrightarrow$ ....

(b) The student blew gently through a straw placed in a test tube containing limewater.



Describe what you would expect to happen to the limewater. Give a reason for your answer. [2]

Observation ....

Reason ....

Examiner only

(c) Quicklime is manufactured from limestone in a lime kiln. A manufacturer expected to obtain 5.6 tonnes per 10 tonnes of limestone used. The actual mass of quicklime produced was 5.1 tonnes.

Calculate the percentage yield of quicklime for this manufacturing process. [2]

percentage yield = ... %

10